



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

 Website: www.ktunotes.in

The time response of a system is the output of the system as a function of time for a known input.

The time response of a control system is divided into two parts:

- i) Transient response
- ii) Steady-state response

In control system, the transient response is defined as that part of the time response as time becomes very large.

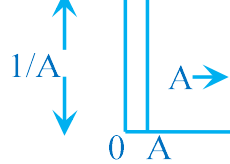
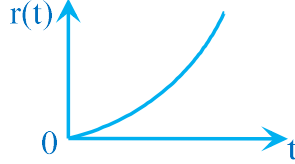
The steady-state response of the system is the response of the system for a given input after the transient has died out.

In steady-state, the output response settles to its final steady-state value or steady-state value.

STANDARD TEST SIGNALS

The inputs to many practical control systems are not exactly known ahead of time. For analysis and design, it is necessary to assume some basic types of test inputs so that a system can be evaluated. Most commonly used test inputs are i) step function, ii) ramp function, iii) parabolic function and iv) impulse function. When the response of a linear time-invariant system is analyzed in the frequency domain, a sinusoidal input with variable frequency is used.

STEP FUNCTION		RAMP FUNCTION	
$r(t) = 0; t < 0$ $r(t) = A; t \geq 0$ $R(s) = \frac{A}{s}$	For unit step, $A = 1;$ $R(s) = \frac{1}{s}$	$r(t) = 0; t < 0$ $r(t) = At; t \geq 0$ $R(s) = \frac{A}{s^2}$	For unit ramp, $A = 1$ $R(s) = \frac{1}{s^2}$

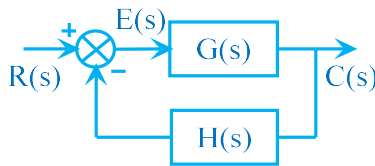


The derivative of a parabolic function is a ramp function

The derivative of a ramp function is a step function

The derivative of a step function is an impulse function

Consider the closed loop system shown in figure below.

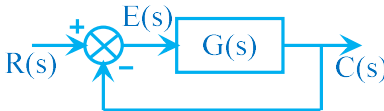


$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

Here, $\frac{C(s)}{R(s)}$ = closed-loop transfer function

$G(s)H(s)$ = open-loop transfer function

For a unity feedback system, $H(s) = 1$.



Here, $c(t)$ is the time response.

TIME-RESPONSE FOR A FIRST ORDER SYSTEM

For a first order system with unity feedback, $G(s) = \frac{1}{sT}$ and $H(s) = 1$.

Closed loop transfer function,
$$\frac{C(s)}{R(s)} = \frac{\frac{1}{sT}}{1 + \frac{1}{sT}} = \frac{1}{1 + sT}$$

$$C(s) = \frac{1}{1 + sT} \times \frac{1}{s}$$

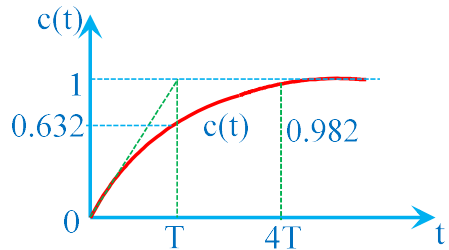
$$C(s) = \frac{\frac{1}{T}}{s(s + \frac{1}{T})} = \frac{A}{s} + \frac{B}{s + \frac{1}{T}} = \frac{1}{s} - \frac{1}{s + \frac{1}{T}}$$

$$c(t) = 1 - e^{-\frac{t}{T}}$$

$$c(0) = 0 \text{ and } c(\infty) = 1$$

At $t = T$, $c(t = T) = 1 - 0.368 = 0.632$; output reaches 63.2% of its final value.

Time constant is defined as the time required for the output response to attain steady-state value.

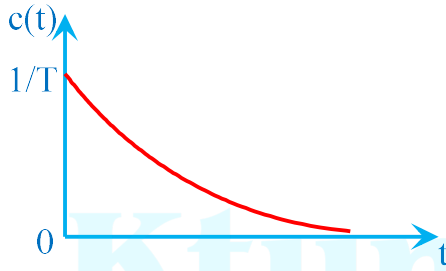


For $t = 4T$, the response reaches around 98.2% of final value. Time required for the response to settle down and stay within a tolerance band of 2% is called settling time T_s . (Sometimes 5%).

Settling time, $T_s = 4T$

Smaller the time constant, the faster the system response.

$$c(t) = \frac{1}{T} e^{-t/T}$$



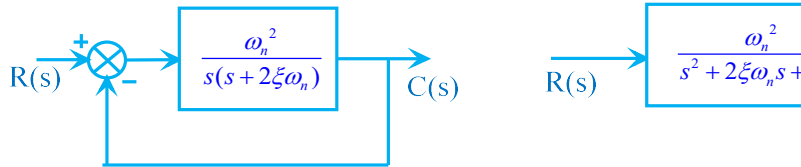
RESPONSE OF A SECOND ORDER SYSTEM

Standard form of closed loop transfer function of a second order system is,

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

ω_n = undamped natural frequency

ξ = damping ratio



Characteristic equation is

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

Roots of characteristic equation will be

$$s_1, s_2 = \frac{-2\xi\omega_n \pm \sqrt{4\xi^2\omega_n^2 - 4\omega_n^2}}{2} = -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$$

$\xi = 0$	$s_1, s_2 = \pm j\omega_n$	Roots are purely imaginary; system is undamped
$\xi = 1$	$s_1, s_2 = -\omega_n$	Roots are real and equal; system is critically damped
$\xi > 1$	$s_1, s_2 = -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$	Roots are real and unequal; system is overdamped

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\xi\omega_n s + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$A = sC(s)\Big|_{s=0} = 1$$

$$\omega_n^2 = (s^2 + 2\xi\omega_n s + \omega_n^2) + Bs^2 + Cs$$

$$\omega_n^2 = (1 + B)s^2 + (2\xi\omega_n + C)s + \omega_n^2$$

$$1 + B = 0 \quad B = -1$$

$$2\xi\omega_n + C = 0 \quad C = -2\xi\omega_n$$

$$C(s) = \frac{1}{s} - \frac{s + 2\xi\omega_n}{s^2 + 2\xi\omega_n s + \omega_n^2} = \frac{1}{s} - \frac{s + 2\xi\omega_n}{(s + \xi\omega_n)^2 - \xi^2\omega_n^2 + \omega_n^2}$$

$$= \frac{1}{s} - \frac{s + 2\xi\omega_n}{(s + \xi\omega_n)^2 + \omega_n^2(1 - \xi^2)}$$

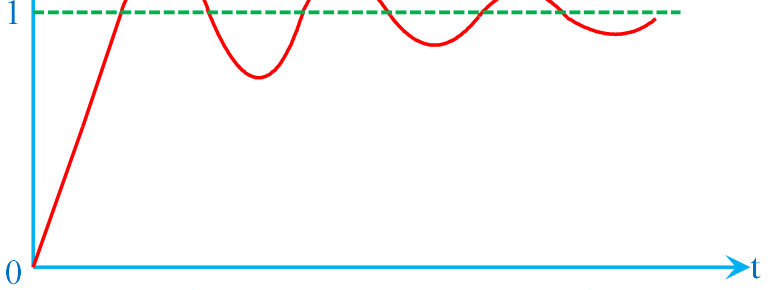
$$= \frac{1}{s} - \frac{s + \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} - \frac{\xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2}$$

$$= \frac{1}{s} - \frac{s + \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} - \frac{\xi\omega_n}{\omega_d} \frac{\omega_d}{(s + \xi\omega_n)^2 + \omega_d^2}$$

$$= \frac{1}{s} - \frac{s + \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} - \frac{\xi}{\sqrt{1 - \xi^2}} \frac{\omega_d}{(s + \xi\omega_n)^2 + \omega_d^2}$$

$$c(t) = 1 - e^{-\xi\omega_n t} \cos \omega_d t - \frac{\xi}{\sqrt{1 - \xi^2}} e^{-\xi\omega_n t} \sin \omega_d t = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1 - \xi^2}} (\sqrt{1 - \xi^2} \cos \omega_d t - \xi \sin \omega_d t)$$

$$\text{Let } \cos \theta = \xi; \sin \theta = \sqrt{1 - \xi^2}$$



UNIT STEP RESPONSE OF CRITICALLY-DAMPED SECOND ORDER

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \omega_n)^2}; \quad \xi = 1$$

For unit step input, $R(s) = \frac{1}{s}$.

$$C(s) = \frac{\omega_n^2}{s(s + \omega_n)^2} = \frac{A}{s} + \frac{B}{(s + \omega_n)^2} + \frac{C}{s + \omega_n}$$

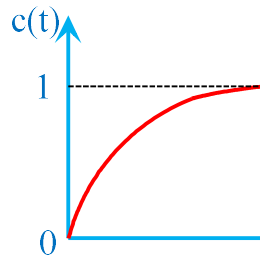
$$A = sC(s)\big|_{s=0} = 1$$

$$B = (s + \omega_n)^2 C(s)\big|_{s=-\omega_n} = -\omega_n$$

$$C = \frac{d}{ds}(s + \omega_n)^2 C(s)\big|_{s=-\omega_n} = \frac{d}{ds} \frac{\omega_n^2}{s}\bigg|_{s=-\omega_n} = \frac{-\omega_n^2}{s^2}\bigg|_{s=-\omega_n} = -1$$

$$C(s) = \frac{\omega_n^2}{s(s + \omega_n)^2} = \frac{1}{s} - \frac{\omega_n}{(s + \omega_n)^2} - \frac{1}{s + \omega_n}$$

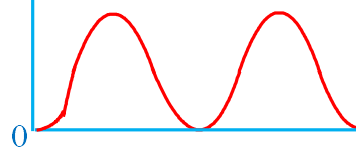
$$c(t) = 1 - \omega_n t e^{-\omega_n t} - e^{-\omega_n t}$$



$$C = 0$$

$$C(s) = \frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{1}{s} - \frac{s}{s^2 + \omega_n^2}$$

$$c(t) = 1 - \cos \omega_n t$$



This response does not die out with time.

UNIT STEP RESPONSE OF OVERDAMPED SECOND ORDER SYSTEM

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}; \xi > 1$$

$$\text{Roots are, } s_1, s_2 = -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$$

$$C(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} R(s)$$

$$\text{For unit step input, } R(s) = \frac{1}{s}.$$

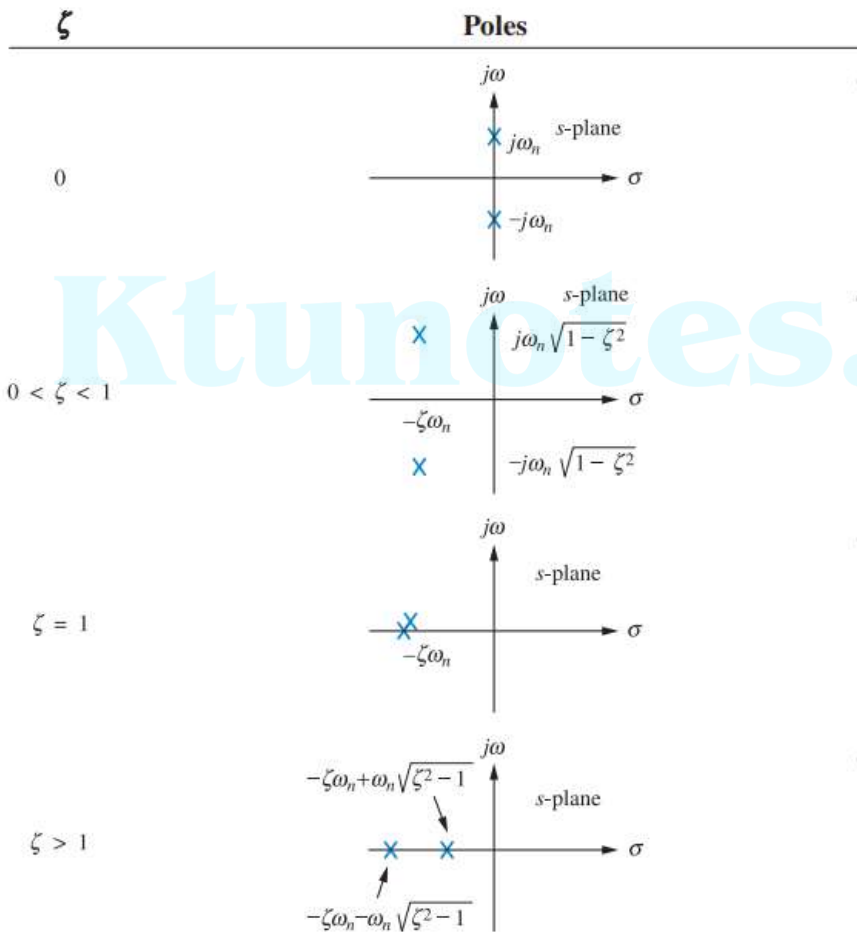
$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\xi\omega_n s + \omega_n^2)} = \frac{\omega_n^2}{s(s - s_1)(s - s_2)} = \frac{A}{s} + \frac{B}{s - s_1} + \frac{C}{s - s_2}$$

$$A = \frac{\omega_n^2}{s_1 s_2} = 1 \quad \text{since } s_1 s_2 = \xi^2 \omega_n^2 - \omega_n^2 (\xi^2 - 1)$$

$$B = \frac{\omega_n^2}{s_1(s_1 - s_2)} = \frac{\omega_n}{2\sqrt{\xi^2 - 1}} \times \frac{1}{s_1} \quad \text{since } s_1 - s_2 = 2\omega_n\sqrt{\xi^2 - 1}$$

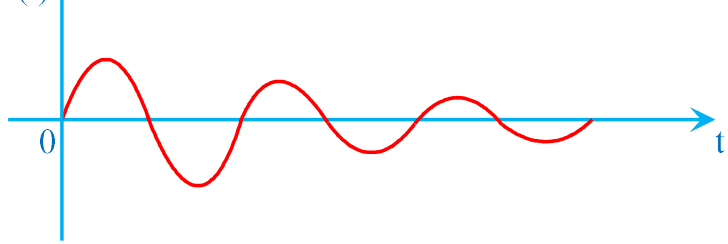
$$C = \frac{\omega_n^2}{s_2(s_2 - s_1)} = \frac{-\omega_n}{2\sqrt{\xi^2 - 1}} \times \frac{1}{s_2}$$

$$C(s) = \frac{1}{s} + \frac{\omega_n}{2\sqrt{\xi^2 - 1}} \left(\frac{1}{s_1(s - s_1)} - \frac{1}{s_2(s - s_2)} \right)$$



UNIT IMPULSE RESPONSE OF UNDERDAMPED SECOND ORDER S

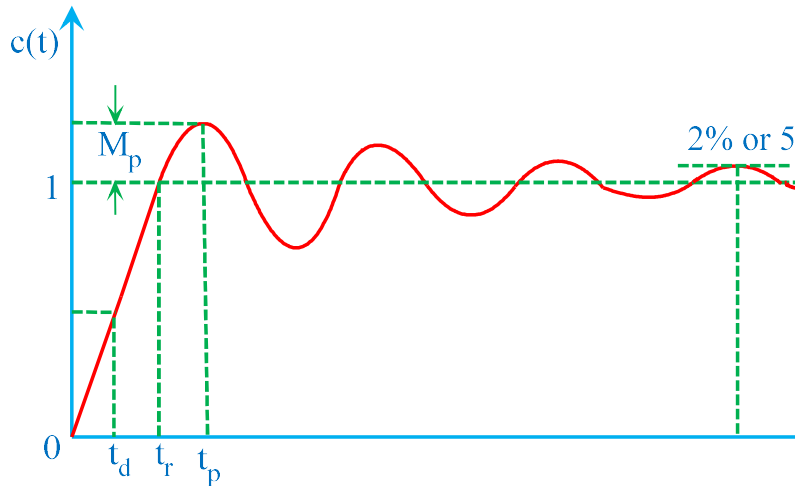
$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}; \quad 0 < \zeta < 1$$



TIME RESPONSE SPECIFICATIONS OF SECOND ORDER SYSTEMS

The performance of a system in time domain is usually evaluated in terms of the

1. How fast it is able to respond to the input?
2. How fast it is reaching the desired output?
3. What is the error between the desired output and the actual output, once and steady state is achieved?
4. Does it oscillate around the desired value?
5. Is the output continuously increasing with time or is it bounded?
6. Our system is Stable or not?



Settling time t_s	The time required for the response to reach and stay within a specified range (2% to 5%) of its final value.
Peak overshoot	Normalised difference between the peak of time output. $\%M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100$
Steady-state error	Difference between the actual output and the desired output. $e_{ss} = \lim_{t \rightarrow \infty} (r(t) - c(t))$

RISE TIME

$$c(t_r) = 1 - \frac{e^{-\xi\omega_n t_r}}{\sqrt{1-\xi^2}} \sin(\omega_d t_r + \theta) = 1$$

$$\sin(\omega_d t_r + \theta) = 0$$

$$\omega_d t_r + \theta = \pi$$

$$t_r = \frac{\pi - \theta}{\omega_d}; \quad \omega_d = \omega_n \sqrt{1 - \xi^2}; \quad \cos \theta = \xi$$

PEAK TIME

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \theta)$$

When $c(t)$ is maximum, $\frac{dc(t)}{dt} = 0$

$$\frac{e^{-\xi\omega_n t_p}}{\sqrt{1-\xi^2}} \cos(\omega_d t_p + \theta) \times \omega_d - \sin(\omega_d t_p + \theta) \times \frac{\xi\omega_n}{\sqrt{1-\xi^2}} \times e^{-\xi\omega_n t_p} = 0$$

$$\omega_d \cos(\omega_d t_p + \theta) - \xi\omega_n \sin(\omega_d t_p + \theta) = 0$$

$$c(t_p) = 1 - \frac{e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin(\omega_d t_p + \theta)$$

$$= 1 - \frac{e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin(\pi + \theta)$$

$$= 1 + \frac{e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin \theta = 1 + \frac{e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \times \sqrt{1-\xi^2} = 1 + e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$$

$$\%M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100 = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} \times 100$$

SETTLING TIME T_s

For 2% tolerance,

$$\frac{e^{-\xi\omega_n t_s}}{\sqrt{1-\xi^2}} = 0.02$$

$$e^{-\xi\omega_n t_s} = 0.02$$

$$-\xi\omega_n t_s = -4$$

$$t_s = \frac{4}{\xi\omega_n} = 4T \text{ where } T = \frac{1}{\xi\omega_n} = \text{time constant}$$

For 5% tolerance,

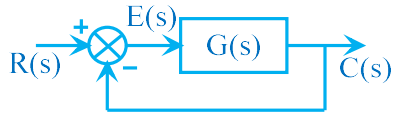
$$\frac{e^{-\xi\omega_n t_s}}{\sqrt{1-\xi^2}} = 0.05$$

$$e^{-\xi\omega_n t_s} = 0.05$$

$$-\xi\omega_n t_s = -3$$

Peak time	$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\xi^2}}$
Peak overshoot	$\%M_p = e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}} \times 100$
Settling time for 2% error	$t_s = \frac{4}{\xi\omega_n}$
Settling time for 5% error	$t_s = \frac{3}{\xi\omega_n}$
Delay time	$t_d = \frac{1+0.7\xi}{\omega_n}$

Obtain the response of unity feedback system whose open loop transfer function is $G(s) = \frac{4}{s^2 + 5s + 4}$ and the input is unit step.



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{4}{s^2 + 5s + 4} = \frac{4}{(s+1)(s+4)}$$

$$C(s) = \frac{4}{s(s+1)(s+4)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+4}$$

$$C(s) = \frac{1}{s} - \frac{4}{3} \times \frac{1}{s+1} + \frac{1}{3} \times \frac{1}{s+4}$$

$$c(t) = 1 - \frac{4}{3}e^{-t} + \frac{1}{3}e^{-4t}$$

Since $0 < \xi < 1$, system is under damped.

The closed loop transfer function of a second order unity feedback system is $\frac{C(s)}{R(s)} = \frac{2}{s^2 + 4s + 2}$. Determine the type of the damping in the system.

$$\frac{C(s)}{R(s)} = \frac{2}{s^2 + 4s + 2}$$

Standard form of second order system is $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$.

Hence, $\omega_n^2 = 2$ and $2\xi\omega_n = 4$

$$\omega_n = 1.414 \text{ rad / s}$$

$$\xi = \frac{4}{2\omega_n} = \frac{4}{2 \times 1.414} = 1.41$$

Since $\xi > 1$, system is over damped.

The closed loop transfer function of a second order unity feedback system is $\frac{C(s)}{R(s)} = \frac{2}{s^2 + 2s + 1}$. Determine the type of the damping in the system.

$$\frac{C(s)}{R(s)} = \frac{2}{s^2 + 2s + 1}$$

Standard form of second order system is $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$. (ω_n^2 in denominator only)

Hence, $\omega_n^2 = 1$ and $2\xi\omega_n = 2$

$$\omega_n = 1 \text{ rad / s}$$

Since $\zeta = 0$, system is undamped.

The open-loop transfer function of a unity feedback system is $G(s) = \frac{4}{s(s+1)}$
time, peak time, peak overshoot and settling time.

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{4}{s^2 + s + 4}$$

Standard form of second order system is $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$.

Hence, $\omega_n^2 = 4$ and $2\xi\omega_n = 1$

$$\omega_n = 2 \text{ rad / s}$$

$$\xi = \frac{4}{2\omega_n} = \frac{4}{2 \times 2} = 0.25$$

$$\omega_d = \omega_n \sqrt{1 - \xi^2} = 2 \times \sqrt{1 - 0.25^2} = 1.936 \text{ rad / s}$$

$$\theta = \cos^{-1} \xi = \cos^{-1} 0.25 = 1.31 \text{ rad}$$

$$\text{Rise time, } t_r = \frac{\pi - \theta}{\omega_d} = \frac{\pi - 1.31}{1.936} = 0.945 \text{ s}$$

$$\text{Peak time, } t_p = \frac{\pi}{\omega_d} = \frac{\pi}{1.936} = 1.622 \text{ s}$$

$$\% \text{ Maximum overshoot, } \%M_p = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} \times 100 = e^{-\frac{\pi \times 0.25}{\sqrt{1-0.25^2}}} \times 100 = 43.26\%$$

$$\text{Settling time for 2\% error, } t_s = \frac{4}{\xi\omega_n} = \frac{4}{0.25 \times 2} = 8 \text{ s}$$

$$\omega_n = 2 \text{ rad / s}$$

$$\xi = \frac{4}{2\omega_n} = \frac{4}{2 \times 2} = 0.25$$

$$\omega_d = \omega_n \sqrt{1 - \xi^2} = 2 \times \sqrt{1 - 0.25^2} = 1.936 \text{ rad / s}$$

$$\theta = \cos^{-1} \xi = \cos^{-1} 0.25 = 1.31 \text{ rad}$$

$$\text{Rise time, } t_r = \frac{\pi - \theta}{\omega_d} = \frac{\pi - 1.31}{1.936} = 0.945 \text{ s}$$

$$\text{Peak time, } t_p = \frac{\pi}{\omega_d} = \frac{\pi}{1.936} = 1.622 \text{ s}$$

$$\% \text{ Maximum overshoot, } \%M_p = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} \times 100 = e^{-\frac{\pi \times 0.25}{\sqrt{1-0.25^2}}} \times 100 = 43.26\%$$

$$\text{Settling time for 2\% error, } t_s = \frac{4}{\xi\omega_n} = \frac{4}{0.25 \times 2} = 8 \text{ s}$$

$$\text{Settling time for 5\% error, } t_s = \frac{3}{\xi\omega_n} = \frac{3}{0.25 \times 2} = 6 \text{ s}$$

A unity-feedback second order system has $\omega_n = 4 \text{ rad / s}$ and $\xi = 0.7$ subject to a unit step input. Determine the open loop transfer function.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} = \frac{16}{s^2 + 5.6s + 16}$$

The open-loop transfer function is

$$G(s) = \frac{\omega_n^2}{s(s + 2\xi\omega_n)} = \frac{16}{s(s + 5.6)}$$

$$\xi = \frac{1}{2\omega_n T} = \frac{1}{2\sqrt{KT}}$$

$$\text{Peak overshoot, } \%M_p = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} \times 100$$

For 85% peak overshoot,

$$e^{-\frac{\xi_1\pi}{\sqrt{1-\xi_1^2}}} = 0.85$$

$$-\frac{\xi_1\pi}{\sqrt{1-\xi_1^2}} = -0.1625$$

$$\xi_1 = 0.0515$$

For 35% peak overshoot,

$$e^{-\frac{\xi_2\pi}{\sqrt{1-\xi_2^2}}} = 0.35$$

$$-\frac{\xi_2\pi}{\sqrt{1-\xi_2^2}} = -1.05$$

$$\xi_2 = 0.317$$

$$\frac{\xi_1}{\xi_2} = \sqrt{\frac{K_2}{K_1}}$$

$$K_2 = 0.0264K_1$$

STEADY STATE ERROR

A steady-state error is defined as the difference between the desired value and system when the response has reached the steady state. It is a measure of system

$$= R(s) - \frac{R(s)G(s)H(s)}{1 + G(s)H(s)}$$

$$= \frac{R(s)[1 + G(s)H(s)] - R(s)G(s)H(s)}{1 + G(s)H(s)}$$

$$= \frac{R(s)}{1 + G(s)H(s)}$$

$$\text{Steady-state error, } e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) = e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)H(s)}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)H(s)}$$

$$\text{For step input, } R(s) = \frac{1}{s}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{1}{1 + G(s)H(s)} = \frac{1}{\lim_{s \rightarrow 0} G(s)H(s)} = \frac{1}{1 + K_p} \text{ where } K_p = \text{static position}$$

$$K_p = \lim_{s \rightarrow 0} G(s)H(s)$$

$$e_{ss} = \frac{1}{1 + K_p}$$

For type 0 system, $K_p = \text{constant}$;

$$e_{ss} = \frac{1}{1 + K_p}$$

For type 1 system, $K_p = \infty$; $e_{ss} = 0$

For type 2 system, $K_p = \infty$; $e_{ss} = 0$

Note: General representation of the transfer function $G(s)H(s)$ of a feedback system is

$$G(s)H(s) = \frac{K(1 + sT_{z1})(1 + sT_{z2}) \dots (1 + sT_{zn})}{s^N(1 + sT_{p1})(1 + sT_{p2}) \dots (1 + sT_{pn})}$$

Here, N is the number of poles at the origin. N represents the **type** number of the system.

For type 1 and above; for unit step input, $e_{ss} = 0$

$$\text{For ramp input, } R(s) = \frac{1}{s^2}$$

$$\lim_{s \rightarrow 0} s^2 [1 + G(s)H(s)] \xrightarrow{s \rightarrow 0} \lim_{s \rightarrow 0} s^2 G(s)H(s) = \frac{Lt}{s \rightarrow 0} s^2 G(s)H(s) = K_a$$

error constant

For type 0 system, $K_a = 0$; $e_{ss} = \infty$

For type 1 system, $K_a = 0$; $e_{ss} = \infty$

For type 2 system, $K_a = \text{constant}$; $e_{ss} = \text{constant}$

For type 3 system, $K_a = \infty$; $e_{ss} = 0$

For type 3 and above; for unit parabolic input, $e_{ss} = 0$

Type	steady-state error e_{ss}		
	unit step	unit ramp	unit parabolic
	$K_p = \lim_{s \rightarrow 0} t G(s)H(s)$	$K_v = \lim_{s \rightarrow 0} t s G(s)H(s)$	$K_a = \lim_{s \rightarrow 0} t s^2 G(s)H(s)$
0	$\frac{1}{1 + K_p}$	∞	∞
1	0	$\frac{1}{K_v}$	∞
2	0	0	$\frac{1}{K_a}$
3	0	0	0

DISADVANTAGES OF STATIC ERROR COEFFICIENT METHOD

1. They do not give any information on the steady-state error when the input is one of the three standard test inputs – step, ramp and parabolic.
2. This method does not provide precise value of error. In most of the cases, the error is either zero or infinity.
3. This method is applicable only to stable system.
4. It does not provide variation of error with respect to time which may be required for many purposes.

The open loop transfer function of a control system is $G(s)H(s) = \frac{K}{s(s^3 + 7s^2 + 12s + 6)}$
 type of system b) error constants K_p , K_v , K_a and c) steady-state error for unit p

Ans: $e_{ss} = 6/K$

A unity feedback system has the forward path transfer function $G(s) = \frac{K}{s(5s + 6)}$
 $r(t) = 1 + 6t$ is applied to the system. Determine the minimum value of K if the
 to be less than 0.1.

$$R(s) = \frac{1}{s} + \frac{6}{s^2}$$

$$sR(s) = \frac{1}{s} + \frac{6}{s^2} = 1 + \frac{6}{s} = \frac{s + 6}{s}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \frac{R(s)}{1 + G(s)H(s)}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \frac{s + 6}{s + sG(s)H(s)} = \frac{6}{K}$$

$$e_{ss} = \frac{6}{K} < 0.1$$

$$K > 60$$

Minimum value of $K = 60$

The closed loop transfer function of a unity-feedback control system is $T(s) = \frac{Ks + \beta}{s^2 + \alpha s + \beta}$
 Determine the steady-state error for unit ramp input.

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)} = \frac{Ks + \beta}{s^2 + \alpha s + \beta}$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \frac{1}{s \left(1 + \frac{Ks + \beta}{s^2 + s(\alpha - K)} \right)} = \frac{\alpha - K}{\beta}$$

DYNAMIC ERROR COEFFICIENTS or GENERALIZED ERROR COEFFICIENTS

Static error coefficients do not give any information on the steady-state error with respect to the three standard test inputs – step, ramp and parabolic.

The dynamic error coefficients provide a simple way of estimating error signals and the steady-state error.

For unity feedback system,

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)} = F(s)$$

The function $F(s)$ can be expanded as a power series in s by Taylor's series expansion.

$$F(s) = \frac{E(s)}{R(s)} = C_0 + C_1 s + \frac{C_2}{2!} s^2 + \frac{C_3}{3!} s^3 + \dots$$

$$E(s) = C_0 R(s) + C_1 s R(s) + \frac{C_2}{2!} s^2 R(s) + \frac{C_3}{3!} s^3 R(s) + \dots$$

$$e(t) = C_0 r(t) + C_1 \dot{r}(t) + \frac{C_2}{2!} \ddot{r}(t) + \frac{C_3}{3!} \dddot{r}(t) + \dots$$

Here, $C_0, C_1, C_2, C_3, \dots$ are called dynamic error coefficients or generalized error coefficients.

The generalized error coefficients can be evaluated as follows:

$$C_0 = \lim_{s \rightarrow 0} F(s)$$

$$C_1 = \lim_{s \rightarrow 0} \frac{dF(s)}{ds}$$

$$C_1 = \frac{1}{K_v}$$

$$C_2 = \frac{1}{K_a}$$

The open loop transfer function of a unity feedback control system is $G(s) = \frac{1}{s^2 + 2s + 1}$. Find the steady-state error using the dynamic error coefficient method for input of $r(t) = 2 + 3t + 2t^3$.

$$F(s) = \frac{1}{1 + G(s)H(s)} = \frac{1}{1 + \frac{1}{s^2 + 2s + 1}} = \frac{s^2 + 2s}{s^2 + 2s + 1}$$

$$C_o = \lim_{s \rightarrow 0} s F(s) = \lim_{s \rightarrow 0} s \frac{s^2 + 2s}{s^2 + 2s + 1} = 0$$

$$C_1 = \lim_{s \rightarrow 0} \frac{dF(s)}{ds} = \lim_{s \rightarrow 0} \frac{d}{ds} \left[\frac{s^2 + 2s}{s^2 + 2s + 1} \right] = \lim_{s \rightarrow 0} \left[\frac{(s^2 + 2s + 1)(2s + 2) - (s^2 + 2s)(2s + 2)}{(s^2 + 2s + 1)^2} \right]$$

$$= \lim_{s \rightarrow 0} \left[\frac{2(s + 1)}{(s + 1)^4} \right] = \lim_{s \rightarrow 0} \left[\frac{2}{(s + 1)^3} \right] = 2$$

$$C_2 = \lim_{s \rightarrow 0} \frac{d^2 F(s)}{ds^2} = \lim_{s \rightarrow 0} \frac{d}{ds} \left[\frac{2}{(s + 1)^3} \right] = \lim_{s \rightarrow 0} \frac{-6}{(s + 1)^4} = -6$$

$$C_3 = \lim_{s \rightarrow 0} \frac{d^3 F(s)}{ds^3} = \lim_{s \rightarrow 0} \frac{d}{ds} \left[\frac{-6}{(s + 1)^4} \right] = \lim_{s \rightarrow 0} \frac{24}{(s + 1)^5} = 24$$

$$r(t) = 2 + 3t + 2t^3$$

$$\dot{r}(t) = 3 + 6t^2$$

$$\begin{array}{r}
 1 + 2s + s^2 \Big) 2s + s^2 \qquad (2s - 3s^2 + 4s^3 \\
 \underline{2s + 4s^2 + 2s^3} \\
 -3s^2 - 2s^3 \\
 \underline{-3s^2 - 6s^3 - 3s^4} \\
 4s^3 + 3s^4
 \end{array}$$

$$F(s) = 2s - 3s^2 + 4s^3 + \dots$$

$$\text{But, } F(s) = C_o + C_1s + \frac{C_2}{2!}s^2 + \frac{C_3}{3!}s^3 + \dots$$

Comparing,

$$C_o = 0$$

$$C_1 = 2$$

$$C_2 = -6$$

$$C_3 = 24$$

$$r(t) = 2 + 3t + 2t^3$$

$$\dot{r}(t) = 3 + 6t^2$$

$$\ddot{r}(t) = 12t$$

$$\ddot{\ddot{r}}(t) = 12$$

$$e(t) = C_o r(t) + C_1 \dot{r}(t) + \frac{C_2}{2!} \ddot{r}(t) + \frac{C_3}{3!} \ddot{\ddot{r}}(t) + \dots$$

$$e(t) = 2(3 + 6t^2) - 3 \times 12t + 4 \times 12 = 54 - 36t + 12t^2$$

The open loop transfer function of a unity feedback control system is $G(s) =$

dynamic error and steady-state error using the dynamic error coefficient

$$r(t) = 1 + 2t + t^2.$$

$$r(t) = 1 + 2t + t^2$$

$$\frac{-s^2 - 0.2s^3}{-s^2 - 0.2s^3 - 0.02s^4} + 0.02s^4$$

$$F(s) = 0.2s - 0.02s^2$$

$$F(s) = C_o + C_1s + \frac{C_2}{2!}s^2 + \frac{C_3}{3!}s^3 + \dots$$

Comparing,

$$C_o = 0$$

$$C_1 = 0.2$$

$$C_2 = -0.04$$

$$C_3 = 0$$

$$e(t) = 0.4(1+t) - 0.08$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \infty$$

A unity feedback system is characterized by the open loop

$$G(s) = \frac{1}{s(0.5s+1)(0.2s+1)}. \text{ Determine the steady state errors for unit-step}$$

acceleration inputs.

$$\text{Ans: } e_{ss} = \frac{1}{1+K_p} = 0$$

$$e_{ss} = \frac{1}{K_v} = 1$$

$$e_{ss} = \frac{1}{K_a} = \infty$$

POLE DOMINANCE FOR HIGHER ORDER SYSTEMS

For a standard 2nd order system, closed loop transfer function is given by $\frac{C(s)}{R(s)}$

The formulas describing percent overshoot, settling time, and peak time were derived for a system with two complex poles and no zeros. If a system has more than two poles or zeros, we can use the formulas to calculate the performance specifications that we derived. In some conditions, a system with more than two poles or with zeros can be approximated by a system that has just two complex dominant poles. Once we justify this approximation,

time of the first overshoot, such parameters as percent overshoot, settling time generated by the second-order underdamped step response component. Thus, approach that of a pure second-order system.

If α_r is not much greater than $\zeta\omega_n$, the real pole's transient response will not be the peak time or settling time generated by the second-order pair. In this case, t_r is significant, and the system cannot be represented as a second-order system.

Note: If the real pole is five times farther to the left than the dominant poles, we is represented by its dominant second-order pair of poles.

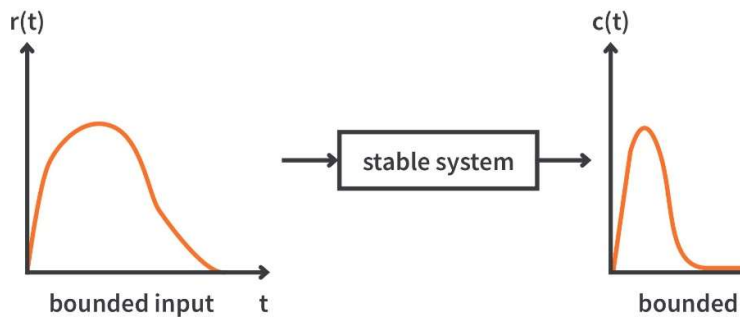
STABILITY

Stability of a system implies that small changes in input do not result in large response.

In an unstable system, any perturbation, no matter how small, will make $y(t)$ unbounded or until the system disintegrates or saturates.

The response of a LTI system is composed of response to initial conditions and

BIBO Stability. A system is BIBO (bounded-input bounded-output) stable if it produces a bounded output. BIBO stability is associated with the response of the state.



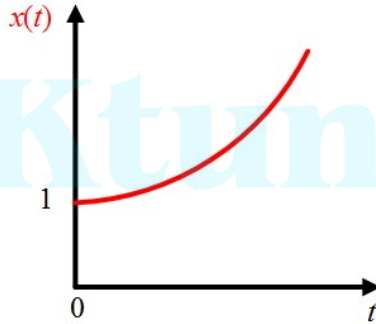
Bounded and unbounded signals:

Unbounded signal has an infinite value at any instant of time.

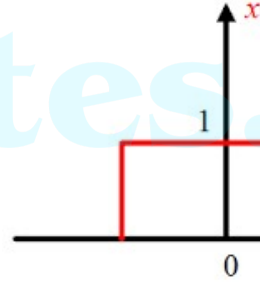
Examples are i) ramp ii) parabolic iii) $\tan(t)$.

Bounded signal has a finite value at any instant of time.

Examples are i) step ii) sinusoidal



(a) unbounded signal



(b) Bounded signal

Asymptotic stability: Asymptotic stability is associated with the response of a system to a bounded input. A system is said to be asymptotically stable if its response to any initial condition decays asymptotically in the steady state.

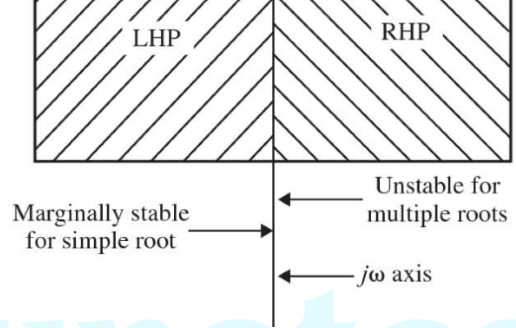
If $y(t)$ is the response due to the initial conditions, the system is asymptotically stable if $y(t) \rightarrow 0$ as $t \rightarrow \infty$.

LOCATION OF ROOTS ON THE S-PLANE FOR STABILITY

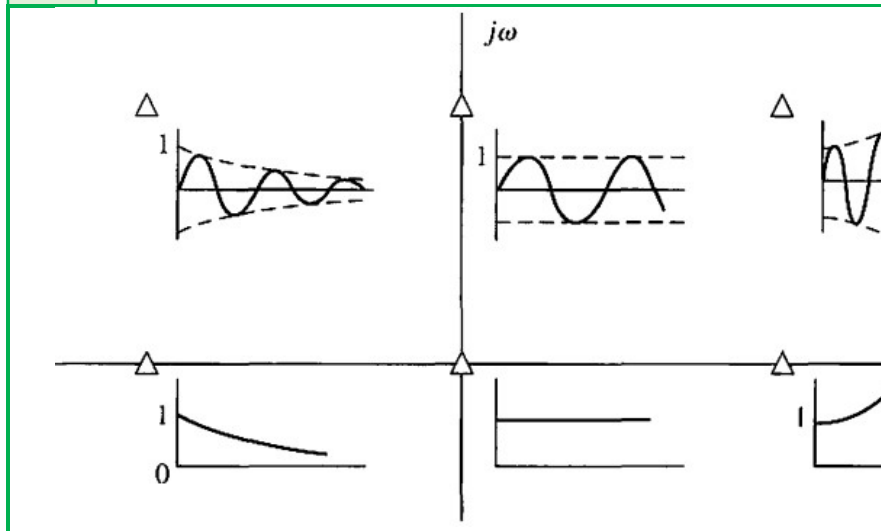
$$\frac{C(s)}{R(s)} = M(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}$$

$$\frac{C(s)}{R(s)} = M(s) = \frac{K (s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)}$$

$z_1, z_2, z_3, \dots, z_m$ = zeros (roots of numerator polynomial)



1	If all the roots of the characteristic equation are on the left-hand side of the s- plane or the roots have negative real parts, then the system is stable.
2	If there is at least one root on the right-hand side of the s- plane or the roots are on the imaginary axis, then the system is unstable and the response bounds with time.
3	If there is at least one root on the imaginary axis, and there are no unstable roots, the response neither decreases to zero, nor increases without bounds. The system is marginally stable.



$$a_0 s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n = 0; \quad a_0 > 0$$

Note: For first and second order systems, positiveness of the coefficients of characteristic equation is a necessary and sufficient condition for stability.

If some of the coefficients are zero or negative, the system is unstable.

If all the coefficients are positive, check for sufficient condition.

ROUTH-HURWITZ STABILITY CRITERION

$$q(s) = a_0 s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n = 0$$

This criterion is based on ordering the coefficients of characteristic equation into Routh array.

$$\begin{array}{c|cccc} s^n & a_0 & a_2 & a_4 & a_6 \\ s^{n-1} & a_1 & a_3 & a_5 & \\ s^{n-2} & b_1 & b_2 & b_3 & \\ \cdot & c_1 & c_2 & & \\ \cdot & & & & \\ s^2 & & & & \\ s^1 & & & & \\ s^0 & a^n & & & \end{array}$$

$$\text{where } b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}; \quad b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1} \dots\dots\dots$$

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}; \quad c_1 = \frac{b_2 a_5 - a_3 b_3}{b_1} \dots\dots\dots$$

For a system to be stable, it is necessary and sufficient that each term of first column be positive if $a_0 > 0$. If this condition is not met, the system is unstable and number of sign changes in the first column corresponds to the number of roots of the characteristic equation in the right half of the s-plane.

$$\begin{array}{c|cc} s^3 & 10 & 5 \\ s^2 & 3.5 & 2 \\ s^1 & -0.714 & 0 \\ s^0 & 2 & \end{array}$$

The elements of the first column have 2 sign changes, hence there are two roots in the right-half of the s-plane and the system is unstable.

Determine the stability of the system whose characteristic equation is $s^5 + 2s^4 + 3s^3 + 4s^2 + 5s + 6 = 0$.

Coefficient of s^4 is missing; hence the system is unstable.

Determine the stability of the system whose characteristic equation is $(s+1)(s^2+2s+2) = 0$.

All the roots of the characteristic equation are on the left-half of the s-plane; hence the system is stable.

SPECIAL CASE 1: First term in any row of the Routh array is zero while rest of the terms are non-zero term.

Determine the stability of the system whose characteristic equation is $s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$.

$$\begin{array}{c|ccc} s^5 & 1 & 2 & 3 \\ s^4 & 1 & 2 & 5 \\ s^3 & 0 & -2 & \\ s^2 & & & \\ s^1 & & & \\ s^0 & & & \end{array}$$

ϵ = small positive number near to zero

$$2\varepsilon + 2$$

Note: We are interested only in the signs of the terms in the first column, not in

Here, there are 2 sign changes, hence there are two roots in the right side of s-plane. The system is unstable.

Determine the stability of the system whose characteristic equation is $s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3 = 0$.

Ans: unstable

SPECIAL CASE 2: All the elements in any one row of the Routh Array are zero.

Determine the stability of the system whose characteristic equation is $s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$.

$$\begin{array}{c|cccc} s^6 & 1 & 8 & 20 & 16 \\ s^5 & 2 & 12 & 16 & \\ s^4 & 2 & 12 & 16 & \\ s^3 & 0 & 0 & 0 & \\ s^2 & & & & \\ s^1 & & & & \\ s^0 & & & & \end{array}$$

$$2s^4 + 12s^2 + 16 = 0$$

$$A(s) = s^4 + 6s^2 + 8 = 0$$

$$\frac{dA(s)}{ds} = 4s^3 + 12s$$

$$s = \pm j\sqrt{2} \text{ and } s = \pm j2$$

Since there are simple roots on imaginary axis, the system is marginally stable.

Note: Roots of auxiliary equation will be the roots of the characteristic equation.

Determine the stability of the system whose characteristic equation is $s^5 + 2s^4 + 2s^3 + 4s^2 + s + 2 = 0$.

$$\begin{array}{c|ccc} s^5 & 1 & 2 & 1 \\ s^4 & 2 & 4 & 2 \\ s^3 & 0 & 0 & \\ s^2 & & & \\ s^1 & & & \\ s^0 & & & \end{array}$$

$$2s^4 + 4s^2 + 2 = 0$$

$$A(s) = s^4 + 2s^2 + 1 = 0$$

$$\frac{dA(s)}{ds} = 4s^3 + 4s = 0$$

$$s^3 + s = 0$$

$$\begin{array}{c|ccc} s^5 & 1 & 2 & 1 \\ s^4 & 2 & 4 & 2 \\ s^3 & 1 & 1 & \\ s^2 & 1 & 1 & \\ s^1 & 0 & 0 & \\ s^0 & & & \end{array}$$

$$A'(s) = s^2 + 1 = 0$$

$$(s^2 + 1)^2 = 0$$

$$s = \pm j, \pm j$$

Since there are repeated roots on imaginary axis, the system is unstable.

The open loop transfer function of unity feedback system is $G(s) = \frac{K}{s(1 + 0.4s)}$
 restriction of K so that the closed loop system is absolutely stable.

$$G(s) = \frac{K}{s(1 + 0.4s)(1 + 0.25s)}$$

Characteristic equation is

$$1 + G(s)H(s) = 0$$

$$s(1 + 0.4s)(1 + 0.25s) + K = 0$$

$$s^3 + 6.5s^2 + 10s + 10K = 0$$

$$\begin{array}{r|rr} s^3 & 1 & 10 \\ s^2 & 6.5 & 10K \\ s^1 & 65 - 10K & \\ s^0 & 6.5 & \\ & 10K & \end{array}$$

For stability, $K > 0$

$$65 - 10K > 0$$

$$65 > 10K$$

$$0 < K < 6.5$$

